An Ideal Linear Secret Sharing Scheme for Complete t-Partite k-Uniform Hypergraph Access Structures

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Abstract

Secret sharing schemes represent a crucial cryptographic protocol, with linear codes serving as a primary tool for their construction. This paper systematically investigates the construction of ideal secret sharing schemes for complete t-partite k-uniform hypergraph access structures using linear codes as the tool. First, it is proved that the generator matrix G of an ideal linear code realizing a complete t-partite 2-uniform hypergraph access structure must have a rank of 2. Simultaneously, a novel method for constructing an ideal secret sharing scheme that realizes such access structures is proposed. Building on this foundation, the case of complete t-partite 2-uniform hypergraphs is extended to complete t-partite k-uniform hypergraphs, and a method for constructing ideal secret sharing schemes to realize them is provided. Compared with existing approaches, both Shamir's method and the scheme proposed by Brickell *et al.* are special cases of our proposed approach.

Index Terms

Ideal secret sharing schemes, linear codes, complete t-partite k-uniform hypergraphs, ideal linear codes

I. Introduction

In 1979, Shamir [19] introduced a threshold secret sharing scheme using polynomial interpolation. That same year, Blakley [4] also presented a threshold sharing scheme based on finite geometry. In 1983, Karnin *et al.* [13] proposed a (k, n) threshold secret sharing scheme based on linear codes. This work unified the approaches of Shamir and Blakley, established both upper and lower bounds on the number of participants, and further extended the discussion to Multisecret Sharing. In 1989, Ito *et al.* [11] proposed a method for constructing perfect secret sharing schemes capable of realizing any access structure, demonstrating that any monotone access structure can be perfectly implemented. However, their scheme suffers from notable efficiency drawbacks, primarily exhibiting a low information rate and undesirably large share sizes. The information rate is a crucial metric for assessing the data expansion of a secret sharing scheme, whose significance is self-evident. In 1996, Beimel [3] established that for any given access structure, a monotone span program can be employed to construct a corresponding linear secret sharing scheme. In 2004, based on Beimel's foundational work, Xiao *et al.* [23] explicitly demonstrated how to construct monotone span programs for realizing linear secret sharing schemes with arbitrary access structures. However, for an arbitrary access structure, finding a secret sharing scheme with the optimal information ratio is highly challenging. In contrast, for graph-based access structures—where the family of authorized sets coincides precisely with the edge set of the graph—researchers can intuitively transform the complex problem of information ratio into a direct analysis of the combinatorial properties of the graph itself, and thereby establish theoretical bounds on its information ratio. In 1991, Brickell *et al.* [6] pointed out that for any graph consisting of disjoint cliques, there exists a corresponding ideal secret sharing scheme whose access structure is determined by the complement of that graph. At the same time, they also provided a construction for ideal schemes on complete multipartite graphs. Specifically, they utilized a threshold scheme to assign identical shares to each clique, and this method achieves the theoretically optimal information rate. In 1992, Brickell *et al.* [7] also provided a construction for ideal secret sharing schemes on complete multipartite graphs. Specifically, they presented a method for constructing ideal schemes for complete multipartite graphs by applying vertex splitting on complete graphs and utilizing orthogonal arrays. Furthermore, by studying the decomposition of graphs, they obtained corresponding decompositions of access structures, thereby enabling the use of schemes for simple access structures to investigate and construct perfect secret sharing schemes for complex access structures. Building on this, they applied and deepened the efficiency metric of information rate, delivering significantly improved theoretical lower bounds for graph-based access structures. In 1995, Blundo *et al.* [5] extended the approach of Brickell *et al.* by decomposing complex graphs into simpler subgraphs—particularly complete multipartite graphs—with the aim of constructing secret sharing schemes with high information rates. Consequently, they derived the

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information rates for a number of graph-based access structures and showed that for any connected graph access structure that is not a complete multipartite graph, the information rate cannot exceed $\frac{2}{3}$. In 2009, Di Crescenzo *et al.* [8] extended the graph decomposition approach of prior work [5] to hypergraph decomposition. They pointed out that any access structure can be represented as a hypergraph, and by decomposing complex hypergraphs into simpler sub-hypergraphs, secret sharing schemes can be constructed. For specific access structures such as hyperpaths and hypercycles, this method yields schemes with improved (or optimal) information rates. Moreover, they indicated that the optimal information rate for these structures is $\frac{2}{3}$ and presented constructions that achieve this bound.

The schemes proposed by Ito *et al*. [11] and the monotone span programs are the primary tools for constructing linear secret sharing schemes. For arbitrary access structures, they proposed two secret sharing schemes. The first scheme is the Disjunctive Normal Form (DNF) secret sharing scheme. From the perspective of minimal authorized sets A_i, each A_i executes an $(|A_i|, |A_i|)$ threshold scheme to share the secret s. In this case, the size of a participant's share depends on how many A_i they belong to. This scheme is efficient when the number of minimal authorized sets is small; otherwise, it is extremely inefficient. The second scheme is the dual of the first, termed the Conjunctive Normal Form (CNF) secret sharing scheme. Starting from maximal unauthorized sets B_i, it assigns each B_i a share r_i of the secret s using the $(|B_i|, |B_i|)$ threshold scheme to assign each B_i a share r_i of the secret s. Thus, for any participant $P \notin B_i$, the held share is $\{r_i\}_{P \notin B_i}$. Here, the size of a participant's share depends on how many B_i they are not in. This scheme is efficient when the number of maximal unauthorized sets is small; conversely, it becomes extremely inefficient otherwise.

Karchmer *et al*. [12] explicitly defined a linear algebraic model to computing monotone Boolean functions, namely the Monotone Span Program (MSP). Specifically, given a field $\mathbb{F}_q$, a matrix M, and a row mapping ρ where each row is labeled by a participant, the matrix $\mathbf{M}_A$ denotes the row vectors labeled by participants in a set A. For a given target vector $\mathbf{v}$, if $\mathbf{v} \in \mathrm{span}\{\mathbf{M}_A\}$, then the MSP is said to accept the set A. Furthermore, if an MSP accepts a set A if and only if $A \in \Gamma$, it is said to accept the access structure Γ. Karchmer also showed that MSPs is equivalent to linear secret sharing schemes. Subsequently, Beimel [3] also proved the correspondence between linear secret sharing schemes and MSPs. Xiao *et al*. [23] further utilized MSPs to construct linear secret sharing schemes implementing general access structures, though this approach is universal yet inefficient.

Linear codes are also crucial tools for constructing linear secret sharing schemes. In 1981, McEliece *et al.* [15] first illustrated the relationship between Reed-Solomon codes (R-S codes) and Shamir's linear secret sharing scheme. In 1993, Massey [14] proposed a secret sharing scheme based on linear codes, demonstrating that the minimal authorized sets in secret sharing schemes correspond one-to-one with minimal codewords in dual codes. For general linear codes, determining all minimal codewords is challenging. In 1998, Ashikhmin and Barg [1] established a sufficient condition for a linear code to be minimal through their investigation into the properties of minimal vectors in general linear codes. In 2003, Ding *et al.* [10] constructed several classes of linear codes (*e.g.*, irreducible cyclic codes and quadratic form codes) whose covering structures can be determined, and utilized them to obtain secret sharing schemes with corresponding access structures. In 2018, Ding *et al.* [9] proposed a necessary and sufficient condition for determining whether a binary linear code is minimal. Based on this condition, they constructed three infinite families of minimal binary linear codes, which break the strict limitation of the classical Ashikhmin–Barg theory and provide new tools for the design of secret sharing schemes. In 2025, Tang *et al.* [21] provided a construction scheme for optimal binary linear codes that realize arbitrary access structures.

In this paper, we focus on constructing ideal secret sharing schemes that realize complete t-partite k-uniform hypergraph access structures using linear codes. The organization of the paper is as follows: Section II introduces the fundamental theoretical knowledge required for the proposed ideal scheme that realizes complete t-partite 2-uniform hypergraph access structures. Section III describes how to identify the adversary structure from a complete t-partite 2-uniform hypergraph access structure. Section IV presents the process of constructing an ideal secret sharing scheme for such access structures using linear codes. Section V generalizes the construction from complete t-partite 2-uniform hypergraphs to complete t-partite k-uniform hypergraphs and provides an ideal secret sharing scheme for the corresponding access structures. Section VI analyzes the advantages of our proposed scheme. Finally, Section VII summarizes the paper and discusses future research directions.

II. Preliminaries

A. Access Structure

Let $P = \{P_1, P_2, \ldots, P_n\}$ denote a set of n participants. Let $AS \subseteq 2^P$ be a nonempty subset of 2^P, where 2^P represents the set of all subsets of the participant set P. That is, AS is a nonempty set composed of specific subsets of P. If AS satisfies the following monotonicity property, we call AS an access structure on P:

- If a set $A \in AS$, then for any subset $A' \subseteq A$ with $A' \in 2^P$, it follows that $A' \in AS$.

For any set in AS, we refer to it as an authorized set over P. Then, AS satisfies monotonicity if any set containing an authorized set is itself an authorized set.

Specifically, if $\Gamma_{\min} = \{A \in \mathrm{AS} \mid \forall B \in 2^P, B \subsetneq A \Rightarrow B \notin \mathrm{AS}\}$, we refer to $\Gamma_{\min}$ as the minimal access structure over P. The elements of $\Gamma_{\min}$ constitute minimal authorized sets—that is, they are authorized sets such that removing any single participant renders the set non-authorizing. In secret sharing schemes, minimal authorized sets define the lowest threshold for secret reconstruction. They eliminate redundant participants, and all authorized sets can be generated from all minimal authorized sets by taking supersets. Possessing a minimal authorized set is equivalent to possessing the access structure.

Therefore, in this paper, the minimal access structure $\Gamma_{\min}$ over P is referred to simply as the access structure $\Gamma_{\min}$, provided this does not cause misunderstanding.

B. Complete t-Partite k-Uniform Hypergraph

Let $V_1, V_2, \ldots, V_t$ be pairwise disjoint vertex sets with $|V_i| = n_i$. If a hypergraph $G = (V, E)$ satisfies:

- **Vertex set:** $V = V_1 \cup V_2 \cup \cdots \cup V_t$, and $|V| = \sum_{i=1}^{t} |V_i| = n$.
- **Edge set:** E consists of all k-element subsets that satisfy:

$$E = \left\{ \{v_1, v_2, \ldots, v_k\} \subseteq V \;\middle|\; v_j \in V_{i_j},\ i_j \in \{1, \ldots, t\},\ \text{and } i_1, i_2, \ldots, i_k \text{ are pairwise distinct} \right\},$$

that is, each hyperedge contains exactly k vertices and each vertex comes from a distinct vertex set, where $2 \le k \le t$,

then G is called a complete t-partite k-uniform hypergraph.

Remark 1. *When $k = 2$, this structure coincides with the classical complete multipartite graph. To maintain terminological consistency and facilitate generalization, this paper uniformly adopts the term "complete t-partite k-uniform hypergraph".*

C. Complete t-Partite k-Uniform Hypergraph Access Structure

Let $P = \{P_1, P_2, \ldots, P_n\}$ denote the set of participants . Let $\Gamma_{\min}$ denote the set of all minimal authorized sets A_i. We represent this access structure graphically as follows:

- Let $G = (V, E)$, where the vertex set $V = P$. For each $E_i \in E$, $E_i \in E$ if and only if $E_i \in \Gamma_{\min}$.

Therefore, the access structure represented by this graph is referred to as a graph access structure. Furthermore, suppose that this graph is a complete t-partite k-uniform hypergraph. In that case, it indicates that the access structure $\Gamma_{\min}$ satisfies the complete t-partite k-uniform hypergraph, abbreviated as the complete t-partite k-uniform hypergraph access structure $\Gamma_{\min}$.

Example 1. $V = \{P_1, \ldots, P_5\} = V_1 \cup V_2 \cup V_3 = \{P_1\} \cup \{P_2, P_3\} \cup \{P_4, P_5\}$, $\Gamma_{\min} = \{\{P_i, P_j\} | P_i \in V_i, P_j \in V_j, i \neq j\}$. *It satisfies the complete 3-partite 2-uniform hypergraph, as shown in Figure 1.*

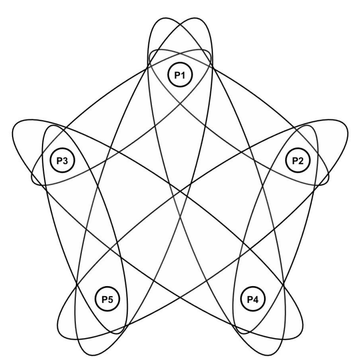

Fig. 1: The complete 3-partite 2-uniform hypergraph access structure.

D. Adversary Structure

For any set in $2^P \backslash AS$, *i.e.*, any subset of P not belonging to AS, it is called a non-authorized set over P. Furthermore, for a given non-authorized set B, if $\forall B' \in 2^p, B \subsetneq B'$, then $B \in AS$, B is termed a maximal non-authorized set over P. The set of all maximal unauthorized sets over P is defined as the adversary structure R over P, abbreviated as adversary structure R.

E. Linear Codes

A k-dimensional subspace of $\mathbb{F}_q^n$ is called a q-ary $[n,k]$ linear code C, *i.e.*, a nonempty subset C of $\mathbb{F}_q^n$ is called a q-ary linear code if

$$a\mathbf{c}+\tilde{a}\tilde{\mathbf{c}} \in C, \quad \forall a,\tilde{a} \in \mathbb{F}_q,\ \mathbf{c},\tilde{\mathbf{c}} \in C, \tag{1}$$

where each element $\mathbf{c}=(c_0,c_1,\ldots,c_{n-1}) \in C$ is called a codeword.

Since the linear code C is a k-dimensional subspace of $\mathbb{F}_q^n$, take k linearly independent codewords from $\mathbf{c}$ and form a $k \times n$ matrix G using their row vectors. This matrix G is called the generator matrix of the linear code C. It is usually assumed that the generator matrix G has no columns consisting entirely of zeros.

The dual code of a linear code C is defined as $C^{\perp}=\{c' \in \mathbb{F}_q^n | \forall \mathbf{c} \in C, \langle \mathbf{c},\mathbf{c}' \rangle = 0\}$, where $\langle \mathbf{c},\mathbf{c}' \rangle$ denotes the vector inner product. It is readily apparent that if C is an $[n,k]$ linear code over $\mathbb{F}_q$, then $C^{\perp}$ is an $[n-k,k]$ linear code over $\mathbb{F}_q$.

Let G be the generator matrix of the linear code C. Then H is the generator matrix of the dual code $C^{\perp}$, also known as the parity-check matrix of the linear code C.

Let C be an $[n,k]$ linear code over the finite field $\mathbb{F}_q$, and let $\mathbf{c}=(c_0,c_1,\ldots,c_{n-1}) \in C$ denote an arbitrary codeword; the support of $\mathbf{c}$, denoted supp($\mathbf{c}$), is defined as the set $\{i \in [0,n-1] \mid c_i \neq 0\}$. A codeword $\mathbf{c}$ is said to cover another codeword $\mathbf{v} \in C$ if supp($\mathbf{v}$) $\subseteq$ supp($\mathbf{c}$); a codeword $\mathbf{c}$ is termed a normalized codeword if its first component satisfies $c_0=1$; finally, a minimal codeword is defined as a normalized codeword that does not cover any normalized codeword other than its scalar multiples over $\mathbb{F}_q$.

F. Information Rate

In secret sharing schemes, the information rate of the access structure serves as a crucial metric for measuring data dispersion. Let $P=\{P_1,P_2,\ldots,P_n\}$ denote a given set of n participants, and let $s \in \mathbb{F}_q$ be the secret. Let s_i denote the sub-secret of participant P_i. Denote $|s|$ and $|s_i|$ as the sizes of the secret domain and the sub-secret domain, respectively.

The information rate relative to the participant P_i is defined as

$$\rho_i = \frac{\log|s|}{\log|s_i|}.$$

Then, the information rate for the access structure implementing this secret sharing scheme is defined as follows:

$$\rho = \min\{\rho_i | 1 \leq i \leq n\} = \frac{\log|s|}{\max\{\log|s_i|\,|1 \leq i \leq n\}}.$$

G. Perfect or Ideal Secret Sharing Scheme

Let the secret s be a random variable that takes values from a finite set S. A secret sharing scheme that realizes a access structure $\Gamma_{\min}$ is said to be perfect if it satisfies the following two conditions:

- **Correctness:** Any authorized set $A \in \Gamma_{\min}$ can accurately reconstruct the secret s from its shares, *i.e.*,

$$H(S|S_A)=0,$$

  where $H(\cdot)$ is the entropy function, and S_A represents the set of shares held by the members in A.
- **Perfect Secrecy:** Any unauthorized set $B \notin \Gamma_{\min}$ cannot obtain any information about S from its shares, *i.e.*,

$$H(S|S_B)=H(S),$$

  where S_B denotes the set of shares held by the members in B.

Furthermore, a secret-sharing scheme that realizes $\Gamma_{\min}$ is called ideal if it is perfect and its information rate is 1.

H. Ideal Linear Codes

$\Gamma_{\min}$ is an access structure over $P=\{P_1,P_2,\ldots,P_n\}$. If there exists an $[n+1,k]$ linear code C over $\mathbb{F}_q$ such that the secret sharing scheme constructed from C satisfies:

- The information rate ρ of the access structure in the secret sharing scheme is 1.
- $\Gamma_{\min}(C)=\Gamma_{\min}$.

We call C the ideal linear code realizing $\Gamma_{\min}$.

III. The Adversary Structure R of The Complete t-Partite 2-Uniform Hypergraph Access Structure

For general access structures, if the access structure $\Gamma_{\min}$ is known, computing the adversarial structure R is difficult, and vice versa [22]. However, for complete t-partite 2-uniform hypergraph access structures, we have the following conclusion.

Theorem 1. *The number of maximal unauthorized sets corresponding to the complete t-partite 2-uniform hypergraph access structure is precisely t, and they are $V_1, V_2, \dots, V_t$ respectively.*

Proof: For any $V_i, 1 \leq i \leq t, V_i$ does not contain any minimal authorized sets and is therefore an unauthorized set; If any additional participant P, where $P \in V_j$ and $i \neq j$, is added to V_i, then the set $V_j \cup \{P\}$ contains participants from two distinct vertex sets V_i and V_j. Consequently, it includes a minimal authorized set, meaning $V_j \cup \{P\}$ is an authorized set. Therefore, each vertex set V_i is a maximal non-authorized set.

Suppose that there exists a maximal unauthorized set C such that $C \neq V_i$ for all $i = 1, 2, \dots, k$. By the definition of a maximal unauthorized set, C must be fully contained within some vertex set V_i, as C cannot contain participants from two or more distinct vertex sets. If C is a proper subset of V_i, then there exists a participant $P \in V_i \setminus C$. Consider the set $C \cup \{P\}$; since this set is also entirely contained within V_i, it remains an unauthorized set. This result contradicts the assumption that C is a maximal unauthorized set. Thus, C must coincide with V_i, implying that no other maximal unauthorized sets exist. ■

Example 2. *In example 1, the maximal unauthorized sets for the access structure are exactly three: $V_1 = \{P_1\}$, $V_2 = \{P_2, P_3\}$, and $V_3 = \{P_4, P_5\}$. The union of these three maximal unauthorized sets constitutes the adversary structure corresponding to this access structure.*

IV. An Ideal Secret Sharing Scheme Based on Linear Codes for Complete t-Partite 2-Uniform Hypergraph Access Structures

A. Necessary and Sufficient Conditions for Ideal Linear Secret Sharing Schemes Realizing Complete t-Partite 2-Uniform Hypergraph Access Structures

In a secret sharing scheme based on linear codes, the linear code is characterized by its generator matrix G. Specifically, given an $[n, k]$ linear code $C \subseteq \mathbb{F}_q^n$, take k linearly independent codewords from the code space. Construct a $k \times n$ generator matrix G using these codewords as row vectors. The linear combinations of the row vectors of G then span the entire linear code space.

Whether a set is authorized depends on the linear relationship between the first column g_0 of the generator matrix G and the columns of G corresponding to the participants in that set. Thus, an ideal linear code C realizes the complete t-partite 2-uniform hypergraph access structure $\Gamma_{\min}$ if and only if the following two conditions are satisfied:

(i) For any two distinct vertex sets V_i, V_j with $1 \leq i \neq j \leq t$, and for every minimal authorized set $\{P_a, P_b\}$, there exist coefficients $\mu, \nu \in \mathbb{F}_q^*$ such that
$$g_0 = \mu g_a + \nu g_b.$$

(ii) For any vertex set V_i with $1 \leq i \leq t$, and for the maximal unauthorized set $\{P_{i1}, \dots, P_{i|V_i|}\}$, the following holds for all coefficients $c_1, c_2, \dots, c_{|V_i|} \in \mathbb{F}_q$:
$$g_0 \neq c_1 g_{i1} + c_2 g_{i2} + \cdots + c_{|V_i|} g_{i|V_i|},$$
where g_{ik} denotes the column vector of G corresponding to participant $P_{ik} \in V_i$.

This is equivalent to the existence of a set of subspaces $L_0, L_1, \dots, L_t$ over $\mathbb{F}_q$ satisfying the following conditions:

1) All column vectors corresponding to participants in the same vertex set V_i lie within the same 1-dimensional subspace L_i (where $0 \leq i \leq t$).
2) For any two distinct vertex sets V_i and V_j, the column vectors corresponding to participants in these two sets belong to distinct 1-dimensional subspaces L_i and L_j (where $0 \leq i \neq j \leq t$).
3) $\text{rank}(G) = 2$.

Theorem 2. *Over the finite field $\mathbb{F}_q$ with $q \geq t$, an ideal linear secret sharing scheme realizing the given complete t-partite 2-uniform hypergraph access structure Γ_{min} exists if and only if 1)–3) hold.*

Proof: Assume that over the finite field $\mathbb{F}_q$ with $q \geq t$, there exists an ideal linear secret sharing scheme that realizes the complete t-partite 2-uniform hypergraph access structure $\Gamma_{\min}$. We proceed to verify that Conditions 1)–3) hold.

Proof of Condition 1): For $i = 0$, the condition holds trivially. For $1 \leq i \leq t$, assume for contradiction that there exist two linearly independent column vectors g_a and g_c corresponding to participants in vertex set V_i. Consider an arbitrary vertex set V_j $(j \neq i)$ and a participant $P_b \in V_j$ with associated column vector g_b. By the definition of minimal authorized sets, $\{P_a, P_b\}$ and $\{P_c, P_b\}$ both constitute minimal authorized sets. Thus, there exist coefficients $\kappa, \lambda, \mu, \nu \in \mathbb{F}_q^*$ such that
$$g_0 = \kappa g_a + \lambda g_b,$$
$$g_0 = \mu g_c + \nu g_b.$$

Equating the two expressions for g_0 yields

$$(\lambda - \nu)g_b = \mu g_c - \kappa g_a.$$

Two cases arise:

- If $\lambda = \nu$, then $\mu g_c - \kappa g_a = 0$. Given that g_a and g_c are linearly independent, this implies $\mu = \kappa = 0$, which contradicts $\mu, \kappa \in \mathbb{F}_q^*$.
- If $\lambda \neq \nu$, then g_b can be expressed as a linear combination of g_a and g_c, *i.e.*,

$$g_b = \frac{\mu}{\lambda - \nu} g_c - \frac{\kappa}{\lambda - \nu} g_a.$$

This implies $g_b \in \text{span}\{g_a, g_c\} \subseteq \text{span}\{g_j \mid P_j \in V_i\}$. Since $g_a \in \text{span}\{g_a, g_c\}$, it follows that

$$g_0 = \kappa g_a + \lambda g_b \in \text{span}\{g_j \mid P_j \in V_i\},$$

which contradicts Theorem 1. Then the initial assumption is invalid, and Condition 1) holds.

Proof of Condition 2): For the case where $i = 0$ and $1 \leq j \leq t$, the column vector g_0 and any column vector g_b corresponding to a participant $P_b \in V_j$ are linearly independent, so the condition holds trivially. The same conclusion applies symmetrically when $j = 0$ and $1 \leq i \leq t$.

For $1 \leq i \neq j \leq t$, assume for contradiction that there exist participants $P_a \in V_i$ and $P_b \in V_j$ whose corresponding column vectors satisfy $g_a = \lambda g_b$ for some $\lambda \in \mathbb{F}_q^*$. By the definition of minimal authorized sets, $\{P_a, P_b\}$ forms a minimal authorized set, so there exist nonzero coefficients $\mu, \nu \in \mathbb{F}_q^*$ such that

$$g_0 = \mu g_a + \nu g_b.$$

Substituting $g_a = \lambda g_b$ into the above equation yields

$$g_0 = \mu(\lambda g_b) + \nu g_b = (\mu\lambda + \nu)g_b.$$

This implies that g_0 is a scalar multiple of g_b, which contradicts Theorem 1. Thus, the assumption is invalid, and Condition 2) holds.

Proof of Condition 3): Take arbitrary distinct vertex sets V_i, V_j $(1 \leq i \neq j \leq t)$, and select participants $P_a \in V_i$, $P_b \in V_j$ with corresponding column vectors g_a, g_b respectively. By the definition of minimal authorized sets, $\{P_a, P_b\}$ is a minimal authorized set, so there exist coefficients $\mu, \nu \in \mathbb{F}_q^*$ such that

$$g_0 = \mu g_a + \nu g_b.$$

Next, take an arbitrary vertex set V_k $(k \neq i)$ and a participant $P_c \in V_k$ with corresponding column vector g_c. Similarly, $\{P_a, P_c\}$ is a minimal authorized set, so there exist nonzero coefficients $\alpha, \beta \in \mathbb{F}_q^*$ satisfying:

$$g_0 = \alpha g_a + \beta g_c.$$

Equating the two expressions for g_0 gives:

$$(\mu - \alpha)g_a = -\nu g_b + \beta g_c.$$

We analyze two cases:

- If $\mu = \alpha$, then $-\nu g_b + \beta g_c = 0$, which implies $g_b = \frac{\beta}{\nu} g_c$. This means g_b and g_c are linearly dependent, contradicting Condition 2).
- If $\mu \neq \alpha$, rearranging the equation yields:

$$g_a = \frac{-\nu}{\mu - \alpha} g_b + \frac{\beta}{\mu - \alpha} g_c.$$

  Here, $\frac{-\nu}{\mu-\alpha} \neq 0$ and $\frac{\beta}{\mu-\alpha} \neq 0$.

By Condition 2), g_b and g_c are linearly independent, so $g_a \in \text{span}\{g_b, g_c\}$.

Since V_i and P_a are arbitrarily chosen, all column vectors of the generator matrix G lie in $\text{span}\{g_b, g_c\}$. Moreover, g_b and g_c are linearly independent, so the dimension of $\text{span}\{g_b, g_c\}$ is 2. Thus, the rank of G is exactly 2.

Combining the proofs of Conditions 1)–3), the necessity is fully proven.

Assume that Conditions 1)–3) are satisfied. We proceed to construct an ideal linear secret sharing scheme over $\mathbb{F}_q$ as follows:

- **Generator Matrix Construction:** Select a nonzero vector $g_0 \in L_0$ to serve as the first column of the generator matrix G. For each vertex set V_i $(1 \leq i \leq t)$ and each participant $P_{ij} \in V_i$ $(1 \leq j \leq |V_i|)$, arbitrarily choose a nonzero vector $g_{ij} \in L_i$. By Condition 3), all vectors are 2-dimensional, and the resulting generator matrix G is of size $2 \times (n+1)$, given by

$$G = \begin{pmatrix} g_0 & g_{11} & \cdots & g_{1|V_1|} & \cdots & g_{t1} & \cdots & g_{t|V_t|} \end{pmatrix}.$$

- **Label Mapping Definition:** Define the label mapping ρ that associates each column of G (excluding the first column g_0) with its corresponding vertex set:

$$\rho = \begin{pmatrix} 2 & \cdots & |V_1|+1 & \cdots & \sum_{i=1}^{t-1}|V_i|+1 & \cdots & n+1 \\ V_1 & \cdots & V_1 & \cdots & V_t & \cdots & V_t \end{pmatrix},$$

where the first row denotes the column indices of G and the second row denotes the vertex set to which the corresponding participant belongs.

By construction, the linear code generated by G yields an ideal linear secret sharing scheme that realizes the access structure $\Gamma_{\min}$. This completes the proof of sufficiency.

Combining the proofs of necessity and sufficiency, the theorem holds. ∎

B. Example

The previous subsection IV-A presents a construction method for ideal linear codes tailored to complete t-partite 2-uniform hypergraph access structures, achieving an information rate that reaches the theoretical optimal value of 1 for ideal linear secret sharing schemes. Based on this, this subsection proposes an ideal secret sharing scheme for the access structure shown in Figure 1.

In $\mathbb{F}_3^2$, find four distinct one-dimensional subspaces $L_0, \ldots, L_4$. Without loss of generality, let

$$L_0 = \mathrm{span}\left\{\begin{pmatrix}1\\0\end{pmatrix}\right\}, \qquad L_1 = \mathrm{span}\left\{\begin{pmatrix}0\\1\end{pmatrix}\right\}, \qquad L_2 = \mathrm{span}\left\{\begin{pmatrix}1\\1\end{pmatrix}\right\}, \qquad L_3 = \mathrm{span}\left\{\begin{pmatrix}1\\2\end{pmatrix}\right\}.$$

Select any column vector from L_0 as the first column g_0 of G. For participants in V_i, select any column vector from L_i as its corresponding column vector in the generator matrix G. Thus, the final form of the generator matrix G is as shown below:

$$G = \begin{pmatrix} 1 & 0 & 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 2 & 2 & 1 \end{pmatrix}.$$

The label mapping ρ is defined as

$$\rho = \begin{pmatrix} 2 & 3 & 4 & 5 & 6 \\ V_1 & V_2 & V_2 & V_3 & V_3 \end{pmatrix},$$

where the first row represents the column numbers in G and the second row denotes the corresponding vertex sets.

The linear code corresponding to this generator matrix G is the ideal linear code for realizing this access structure. The verification process is as follows:

Massey [14] demonstrated that minimal codewords in the dual code correspond one-to-one with minimal authorized sets. This paper adopts this method to verify whether the linear code corresponding to generator matrix G is the ideal linear code for this access structure—that is, if all minimal codewords appearing in the dual code precisely correspond to the minimal authorized sets specified herein. Based on coding theory, the parity-check matrix H corresponding to generator matrix G is as follows:

$$H = \begin{pmatrix} 2 & 2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

It is readily apparent that over $\mathbb{F}_3$, the codewords formed by all linear combinations of the row vectors of this parity-check matrix H constitute the entire codeword set of the dual code. Upon verification, the minimal codeword set of the dual code precisely matches the access structure of this example, with no new minimal codewords appearing. This result confirms that the ideal linear code corresponding to the generator matrix G indeed realizes the access structure specified in this example.

V. An Ideal Secret Sharing Scheme Based on Linear Codes for Complete t-Partite k-Uniform Hypergraph Access Structures

In this section, we will discuss ideal secret sharing schemes for realizing arbitrary complete t-partite k-uniform hypergraph access structures, where $2 \leq k \leq t$.

A. Minimal Authorized Sets of Size k

Building upon the discussion of the basic case, this subsection extends the relevant conclusions to the more general case of complete t-partite k-uniform hypergraph access structures for $k > 2$. First, the characterization of the adversary structure in Theorem 1 is generalized to obtain the following general result.

Theorem 3. *The adversary structure R corresponding to the complete t-partite k-uniform hypergraph access structure $\Gamma_{\min}$ is defined as follows:*

$$R = \left\{ \bigcup_{i=j_1}^{j_{k-1}} V_i \,\middle|\, \{j_1, j_2, \ldots, j_{k-1}\} \subseteq \{1, 2, \ldots, t\} \right\}.$$

Proof: Following the same argument as in the proof of Theorem 1, the result holds (the proof is omitted). ■

Correspondingly, the algebraic characterization of the existence of ideal linear codes in Theorem 2 can also be extended to the current general case. Specifically, an ideal linear code that realizes this access structure must satisfy the following necessary and sufficient condition.

Theorem 4. *Over $\mathbb{F}_q$, where $q \geq t$, there exists an ideal linear code C that realizes the complete t-partite k-uniform hypergraph access structure $\Gamma_{\min}$ if and only if the following conditions a)–c) hold:*

a) For any given participant subset $V_i, 0 \leq i \leq t$, all column vectors corresponding to participants within V_i lie in the same 1-dimensional subspace L_i.

b) For any k distinct participant subsets $V_{i_1}, V_{i_2}, \ldots, V_{i_k}$ with $\{i_1, i_2, \ldots, i_k\} \subseteq \{0, 1, \ldots, t\}$, the column vectors corresponding to the participants in each subset lie in distinct 1-dimensional subspaces $L_{i_1}, L_{i_2}, \ldots, L_{i_k}$, and they are linearly independent.

c) $rank\,(G) = k$.

Proof: The necessity follows by the same argument as in the proof of necessity for Theorem 2 (the detailed proof is omitted).

For sufficiency, consider the following subspaces in $\mathbb{F}_q^k$:

$$L_{i_0} = \operatorname{span}\left\{ \begin{pmatrix} 0 \\ \vdots \\ 0 \\ a \end{pmatrix} \right\}, \qquad L_{i_1} = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ x_1^1 \\ \vdots \\ x_1^{k-1} \end{pmatrix} \right\}, \qquad \cdots \qquad L_{i_t} = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ x_t^1 \\ \vdots \\ x_t^{k-1} \end{pmatrix} \right\},$$

where $a \in \mathbb{F}_q^*$ and $x_1, x_2, \ldots, x_t \in \mathbb{F}_q$ are distinct. Let the basis vectors of the above subspaces be $w, v_1, \ldots, v_t$. Consider the set of vectors $\{w, v_1, \ldots, v_t\}$. It is readily apparent that any k vectors selected from this set are linearly independent, while any $k+1$ vectors are linearly dependent. This is guaranteed by the distinctness of $x_1, x_2, \ldots, x_t$ being distinct and the properties of the Vandermonde matrix.

Therefore, based on these $t+1$ one-dimensional subspaces, we can construct an ideal linear code over $\mathbb{F}_q$ that realizes this access structure. ■

B. Examples with Authorized Subsets of Size 3

Let the set of participants be $V = V_1 \cup V_2 \cup V_3 \cup V_4 \cup V_5 = \{P_1, P_2\} \cup \{P_3, P_4, P_5\} \cup \{P_6\} \cup \{P_7\} \cup \{P_8\}$. The complete 5-partite 3-uniform hypergraph access structure $\Gamma_{\min}$ is defined as follows:

$$\begin{aligned}
\Gamma_{\min} = &\big\{\{P_1, P_3, P_6\}, \{P_1, P_4, P_6\}, \{P_1, P_5, P_6\}, \{P_2, P_3, P_6\}, \{P_2, P_4, P_6\}, \{P_2, P_5, P_6\}\big\} \\
&\cup \big\{\{P_1, P_3, P_7\}, \{P_1, P_4, P_7\}, \{P_1, P_5, P_7\}, \{P_2, P_3, P_7\}, \{P_2, P_4, P_7\}, \{P_2, P_5, P_7\}\big\} \\
&\cup \big\{\{P_1, P_3, P_8\}, \{P_1, P_4, P_8\}, \{P_1, P_5, P_8\}, \{P_2, P_3, P_8\}, \{P_2, P_4, P_8\}, \{P_2, P_5, P_8\}\big\} \\
&\cup \big\{\{P_1, P_6, P_7\}, \{P_2, P_6, P_7\}\big\} \\
&\cup \big\{\{P_1, P_6, P_8\}, \{P_2, P_6, P_8\}\big\} \\
&\cup \big\{\{P_1, P_7, P_8\}, \{P_2, P_7, P_8\}\big\} \\
&\cup \big\{\{P_3, P_6, P_7\}, \{P_4, P_6, P_7\}, \{P_5, P_6, P_7\}\big\} \\
&\cup \big\{\{P_3, P_6, P_8\}, \{P_4, P_6, P_8\}, \{P_5, P_6, P_8\}\big\} \\
&\cup \big\{\{P_3, P_7, P_8\}, \{P_4, P_7, P_8\}, \{P_5, P_7, P_8\}\big\} \\
&\cup \big\{\{P_6, P_7, P_8\}\big\}.
\end{aligned}$$

By Theorem 3, the adversary structure corresponding to $\Gamma_{\min}$ is as follows:

$$\begin{aligned} R =& \{V_1 \cup V_2, V_1 \cup V_3, V_1 \cup V_4, V_1 \cup V_5\} \\ & \cup \{V_2 \cup V_3, V_2 \cup V_4, V_2 \cup V_5\} \\ & \cup \{V_3 \cup V_4, V_3 \cup V_5\} \\ & \cup \{V_4 \cup V_5\}. \end{aligned}$$

By Theorem 4, the following subspaces are selected in $\mathbb{F}_5^3$:

$$L_{i_0} = \operatorname{span}\left\{\begin{pmatrix}0\\0\\1\end{pmatrix}\right\}, \qquad L_{i_1} = \operatorname{span}\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}\right\}, \qquad L_{i_2} = \operatorname{span}\left\{\begin{pmatrix}1\\1\\1\end{pmatrix}\right\},$$
$$L_{i_3} = \operatorname{span}\left\{\begin{pmatrix}1\\2\\4\end{pmatrix}\right\}, \qquad L_{i_4} = \operatorname{span}\left\{\begin{pmatrix}1\\3\\4\end{pmatrix}\right\}, \qquad L_{i_5} = \operatorname{span}\left\{\begin{pmatrix}1\\4\\1\end{pmatrix}\right\}.$$

Based on the above subspaces, the generator matrix for an ideal linear code realizing $\Gamma_{\min}$ is given as follows:

$$G = \begin{pmatrix} 0 & 1 & 3 & 2 & 3 & 4 & 3 & 2 & 3 \\ 0 & 0 & 0 & 2 & 3 & 4 & 1 & 1 & 2 \\ 2 & 0 & 0 & 2 & 3 & 4 & 2 & 3 & 3 \end{pmatrix}.$$

The label mapping ρ is defined as

$$\rho = \begin{pmatrix} 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ V_1 & V_1 & V_2 & V_2 & V_2 & V_3 & V_4 & V_5 \end{pmatrix},$$

where the first row represents the column numbers in G and the second row denotes the corresponding vertex sets.

VI. Advantages of Our Scheme

A. Comparison with Monotone Span Programs

This subsection selects the access structure showed in Figure 1 as a test case. We realized this access structure using the proposed scheme in this paper as well as the schemes constructed by Xiao *et al*. [23] and Ito *et al*. [11], followed by comparative experiments. The results demonstrate that compared to these two classical schemes, the proposed method significantly reduces the row and column sizes of the generator matrix while elevating the information rate to the theoretical optimum of 1. This achievement reaches a theoretical upper bound unattainable by the existing schemes in terms of information rate. This outcome fully validates the dual advantages of the proposed method in both resource efficiency and theoretical performance.

1) Scheme of Xiao et al.: Over $\mathbb{F}_3$, given a nonzero 16-dimensional target vector $\mathbf{v} = \begin{pmatrix}1 & 1 & \cdots & 1\end{pmatrix}$, let matrix M is as shown below:

$$M = \begin{pmatrix}
1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}.$$

The label mapping ρ is defined as

$$\rho = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 & 16 \\ P_1 & P_2 & P_1 & P_3 & P_1 & P_4 & P_1 & P_5 & P_2 & P_4 & P_2 & P_5 & P_3 & P_4 & P_3 & P_5 \end{pmatrix},$$

where the first row represents the row indices of the matrix M, and the second row indicates the corresponding participants.

The number of rows of matrix M is termed the row size of the monotone span program M, denoted as $\mathrm{Rsize}(M)$. Correspondingly, the number of columns of M is defined as its column size, denoted as $\mathrm{Csize}(M)$.

Thus, a monotone span program $\mathcal{M}(M,\rho)$ is obtained that computes $f_{\Gamma_{min}}$, where $\mathrm{Rsize}\,(M) = \mathrm{Csize}\,(M) = 16$. This further enables the obtained linear secret sharing scheme to realize this access structure.

2) Scheme of Ito et al.: The following reformulates the secret sharing scheme proposed by Ito *et al.* using a monotone span program. Over $\mathbb{F}_3$, given a nonzero 3-dimensional target vector $v = \begin{pmatrix}1 & 1 & 1\end{pmatrix}$, let matrix M is as shown below:

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

The label mapping ρ is defined as

$$\rho = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ P_2 & P_3 & P_4 & P_5 & P_1 & P_4 & P_5 & P_1 & P_2 & P_3 \end{pmatrix},$$

where the first row represents the row indices of the matrix M, and the second row indicates the corresponding participants.

Thus, a monotone span program $\mathcal{M}(M,\rho)$ is obtained that computes $f_{\Gamma_{min}}$, where $\mathrm{Rsize}\,(M) = 10, \mathrm{Csize}\,(M) = 3$. This further enables the obtained linear secret sharing scheme to realize this access structure.

3) Scheme Comparison: We compare these schemes both in terms of the row and column sizes and the information rate, as shown in Table I and Table II, respectively.

TABLE I: Comparison of Row and Column Sizes for Three Schemes

Schemes	Rsize	Crize
Scheme of Xiao *et al.*	16	16
Scheme of Ito *et al.*	10	3
Our schemes	6	2

TABLE II: Comparison of Information Rates for Three Schemes

Schemes	Information rates
Scheme of Xiao *et al.*	$\frac{1}{4}$
Scheme of Ito *et al.*	$\frac{1}{2}$
Our schemes	1

The two aforementioned classical schemes both fail to achieve the theoretically optimal information rate, and the resulting matrices have a large size in terms of both rows and columns, which to some extent restricts their practical applicability. Through a comparison between our proposed scheme and the two classical schemes, we conclude that our scheme demonstrates significant advantages in terms of the row size, column size, and information rate of the matrix.

B. Comparison with Scheme of Brickell et al. [6]

Brickell *et al.* first proved and presented an ideal secret sharing scheme realizing a complete t-partite 2-uniform hypergraph access structure. Their ideal scheme utilizes a threshold scheme to achieve equal shares for each participant within a clique. In contrast, Theorem 2 proves and presents an ideal secret sharing scheme for a complete t-partite 2-uniform hypergraph access structure from the perspective of linear codes. It further analyzes and reveals that participants within the same clique (referred to as vertex set V_i throughout this paper and subsequent sections) need not necessarily have equal shares. This indicates that the condition of equal shares among participants in each vertex set V_i within scheme of Brickell *et al.* can be relaxed. It suffices that the column vectors of the generator matrix G corresponding to participants within the same vertex set V_i lie in the same one-dimensional space (corresponding to proportional column vectors). This discovery significantly relaxes the construction conditions for ideal schemes and expands the design space for such schemes.

Moreover, we will subsequently show that setting the proportionality coefficient between the column vectors of the generator matrix G corresponding to participants within the same vertex set V_i to 1 leads to a degenerate case that reduces to scheme of Brickell *et al.*. This means that while maintaining an information rate of 1, we have overcome the constraint of equal shares for participants in the same vertex set, establishing a more general and flexible optimal scheme.

C. Comparison with the Reed-Solomon Code for Threshold Access Structures

For the classical (k,t) threshold access structure, over the finite field $\mathbb{F}_q$ where $q \geq t$, there exists a $[t+1,k]$ R-S code realizing this access structure. Its generator matrix G is the following $k \times (t+1)$ Vandermonde matrix:

$$G = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_0^1 & x_1^1 & \cdots & x_t^1 \\ x_0^2 & x_1^2 & \cdots & x_t^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_0^{k-1} & x_1^{k-1} & \cdots & x_t^{k-1} \end{pmatrix},$$

where $x_0 = 0, x_1, \ldots, x_t$ are distinct nonzero elements in $\mathbb{F}_q$.

By Theorem 4, we also know that over a finite field $\mathbb{F}_q$ with $q \geq t$, we can construct an ideal $k \times (t+1)$ generator matrix G for an ideal $[t+1,k]$ linear code corresponding to the complete t-partite k-uniform hypergraph access structure $\Gamma_{\min}$ from the perspective of subspaces.

Therefore, the following conclusion holds: when the participant set P is partitioned into t disjoint subsets, each containing only one participant, this degenerates into the classical (k,t) threshold access structure. It can be realized by an ideal $[t+1,k]$ linear code over a smaller finite field $\mathbb{F}_q$, where $q \geq t$. Moreover, while in the original $[t+1,k]$ R-S code, the generator matrix G assigns each participant P_i the column vector is $v_i = \begin{pmatrix} 1 \\ {x_i}^1 \\ \vdots \\ x_i^{k-1} \end{pmatrix}$, Theorem 4 reveals that in fact, the column vector corresponding to each participant P_i has more choices. That is, the column vector corresponding to P_i can be any column vector in the subspace spanned by $\begin{pmatrix} 1 \\ {x_i}^1 \\ \vdots \\ x_i^{k-1} \end{pmatrix}$.

VII. Conclusion

A. Discussion of Results

This paper systematically investigates the construction of ideal linear secret sharing schemes for the important class of graph access structures known as complete t-partite k-uniform hypergraphs. The research follows a path from specific cases to general theory, combining theoretical analysis with concrete examples. First, using complete t-partite 2-uniform hypergraphs as a special case, their adversary structure is precisely characterized. Necessary and sufficient conditions for linear codes that realize this structure ideally are established, and the feasibility of the scheme is verified through specific examples. Building on this foundation, the theoretical framework is fully extended to the general case $(2 \leq k \leq t)$. Not only is the adversary structure for the general case characterized, but universal necessary and sufficient conditions for ideal linear codes realizing this class of access structures are also established. This is complemented by examples that clearly illustrate the construction process from theoretical conditions to concrete schemes. Finally, the advantages of the proposed scheme are highlighted through multi-faceted comparisons: compared to the monotone span program, it provides an algebraic construction with optimal information rate; compared to the method by Brickell *et al.*, it offers more flexible construction; and compared to the threshold scheme based on Reed-Solomon codes, it successfully addresses the ideal realization problem for non-threshold access structures of this hypergraph type. This study deeply integrates the algebraic methods of linear codes with this class of hypergraph access structures, providing a unified, rigorous, and constructible framework for ideal secret sharing schemes for such structures.

B. Future Work

In the theoretical framework of ideal secret sharing, the work of Brickell and Davenport [6] revealed a profound connection between matroid ports and ideal access structures: on the one hand, if an access structure is ideal, then it must be the port of some matroid (necessity); on the other hand, if an access structure is the port of a matroid representable over a finite field, then it must be ideal (sufficiency). However, a well-known "gap" exists between these two conditions—there are access structures that are ports of matroids but are not ports of any representable matroid. A prime example [18] is the port of the Vámos matroid, which has been proven to be non-ideal. This signifies that "being a matroid port" is only a necessary but not sufficient condition for idealness. Consequently, the complete characterization of ideal access structures in general remains an open problem.

In specific constrained settings, however, this gap vanishes. For instance, Beimel *et al.* [2] proved that for a binary or ternary domain of secrets, an access structure is ideal over that domain if and only if it is the port of a matroid representable over that domain. In this case, the necessary and sufficient conditions are fully equivalent. Furthermore, for many specific families of access structures, such as graph-defined access structures [6] and bipartite access structures [17], research has also demonstrated that their idealness is equivalent to being the port of a representable matroid. This indicates that, despite the theoretical gap in the general case, the Brickell-Davenport theoretical framework provides a complete and clear characterization of idealness for small domains or for access structure families with favorable combinatorial features. Thus, it offers a crucial foundation for understanding ideal secret sharing and for further exploring the representation gap in the general case.

Inspired by this, having demonstrated in our work that the class of graph access structures based on complete t-partite k-uniform hypergraphs can be realized by ideal linear codes over $\mathbb{F}_q$ (where $q \geq t$), a deeper question naturally arises: Could this class of graph access structures be the port of a representable matroid? In other words, for this specific family of access structures, can we close the representability gap in the theory of Brickell and Davenport?

Furthermore, we note that the Non-Pappus matroid is not linearly representable over any field [16, P. 39], yet it can be multilinearly representable. Simonis *et al.* [20] constructed the Non-Pappus matroid as a multilinearly representable matroid and proved that there exists a class of access structures (namely, the ports of this matroid) which can be realized by an ideal multilinear secret sharing scheme, but cannot be realized by any ideal linear secret sharing scheme. This example leads us to the following question: if a complete t-partite k-uniform hypergraph is not the port of a linearly representable matroid, could it possibly be the port of a multilinearly representable matroid? If so, how can one construct the corresponding ideal multilinear secret sharing scheme?

Remark 2. *Multilinear secret sharing is a generalization of linear secret sharing. In a linear scheme, the secret is a single element from a finite field. In a multilinear scheme, the secret is a vector over the field (i.e., composed of multiple elements). Correspondingly, the share of each participant is also a vector.*

Following this line of thinking, we further explore a more general research direction from the perspective of graph structures. Inspired by Blundo *et al.* [5] and Di Crescenzo *et al.* [8], we consider: can any complex hypergraph be decomposed into multiple simpler sub-hypergraphs, in particular, complete t-partite k-uniform hypergraphs? If so, this may enable the construction of secret sharing schemes with favorable information ratios by combining schemes for these simpler substructures, along with further analysis of bounds on their information ratios.

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